Skills Summary

Complex Number Operations

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding and Subtracting

Rule: treat i like a variable and collect like parts.

$(a+bi)+(c+di)=(a+c)+(b+d)i \qquad (a+bi)-(c+di)=(a-c)+(b-d)i$

A subtraction sign applies to *both* parts of the number behind it.

Example: Simplify $(6+2i)-(9-5i)$

Step 1. Nothing here needs multiplying — both numbers are already in $a+bi$ form, so the job is only to distribute the minus sign and collect like parts.

Step 2. $6+2i-9+5i$, so the real parts give $6-9$ and the imaginary parts give $2+5$.

$\Rightarrow \quad \mathbf{-3+7i}$

Try it: Simplify $(4-9i)+(-6+3i)$

check: $\mathbf{-2-6i}$

Skill 2 — Multiplying Two Complex Numbers

Rule: multiply out all four products, then replace i^2 with -1 and collect.

$(a+bi)(c+di)=(ac-bd)+(ad+bc)i$

The $-bd$ in the real part is exactly the $i^2=-1$ step written once and for all.

Example: Multiply $(3+5i)(2-4i)$

Step 1. Two binomials multiplied together — use the same four products as ordinary algebra, then convert the i^2 term.

Step 2. $6-12i+10i-20i^2$, and $-20i^2=+20$, so the real parts give $6+20$ and the imaginary parts give $-12+10$.

$\Rightarrow \quad \mathbf{26-2i}$

Try it: Multiply $(4+i)(3-2i)$

check: $\mathbf{14-5i}$

Skill 3 — Conjugate Pairs and Real Products

Rule: $a + bi$ and $a - bi$ are *conjugates*. Their product is always real:

$(a + bi)(a - bi) = a^2 + b^2$

The two middle terms are opposites, so they cancel; the i^2 term flips sign and *adds* to the square of the real part.

Example: Multiply $(7 + 3i)(7 - 3i)$
Step 1. The factors differ only in the sign of the imaginary part, so this is a conjugate pair — expect the i terms to cancel and the answer to be real.
Step 2. The middle terms are $+21i$ and $-21i$, which add to zero, and what is left is the sum of the squares of the parts.
$\Rightarrow$ **58**

Try it: Multiply $(4 + 5i)(4 - 5i)$

check: **41**

Skill 4 — Squaring a Complex Number

Rule: a square is still a product of two binomials.

$(a + bi)^2 = a^2 - b^2 + 2abi \qquad (a - bi)^2 = a^2 - b^2 - 2abi$

Example: Expand $(5 + 2i)^2$
Step 1. Squaring a binomial is the same product twice, so the middle term must appear — write it out rather than squaring each piece.
Step 2. $25 + 10i + 10i + 4i^2$, and $4i^2 = -4$, so the real parts give $25 - 4$ and the imaginary parts give $10 + 10$.
$\Rightarrow$ $\mathbf{21 + 20i}$

Try it: Expand $(6 - i)^2$

check: $\mathbf{35 - 12i}$

(!) Watch out: $(a + bi)^2$ is *not* $a^2 + b^2 i^2$. Losing the middle term $2abi$ is the most common error on this page. And $i^2 = -1$, never $+1$ — if your real part came out too small, check that sign first.